



MATHEMATICS

0580/22

Paper 2 (Extended)

March 2019

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the March 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **5** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1	-14	1	
2	330	1	
3	$\frac{23}{99}$	1	
4(a)	0.047	1	
4(b)	2.76×10^6	1	
5	467.42 or 467	2	M1 for $500 \div 1.0697$
6	70	2	M1 for $25\,000 \times 0.0028$ oe
7	308	2	M1 for $180 + 128$ oe or 52 seen
8	$x + 7y$	2	M1 for a correct route
9	$[y =]\frac{1}{4}(x - 4)$ oe final answer	2	M1 for $y = k(x - 4)$
10	375	3	M2 for $2(12 \times 5 + 12 \times 7.5 + 5 \times 7.5)$ oe or M1 for 12×5 or 12×7.5 or 5×7.5
11	$22\frac{2}{9}$ or 22.2 or 22.22...	3	M2 for $\frac{77-63}{63}[\times 100]$ oe or $\frac{77}{63} \times 100 [-100]$ oe or M1 for $\frac{77}{63}$ oe
12	4.21 or 4.212....	3	M2 for $\sqrt{\frac{275 \times 3}{14.8 \times \pi}}$ oe or M1 for $275 = \frac{1}{3} \times \pi \times r^2 \times 14.8$ oe

Question	Answer	Marks	Partial Marks
13(a)	$k(7k - 15)$ final answer	1	
13(b)	$4(m + p)(3 + 2m + 2p)$ final answer	2	B1 for $(m + p)(12 + 8(m + p))$ or $(m + p)(12 + 8m + 8p)$ or $(4m + 4p)(3 + 2m + 2p)$ or $(2m + 2p)(6 + 4m + 4p)$ or $2(2m + 2p)(3 + 2m + 2p)$ or $2(m + p)(6 + 4m + 4p)$
14	6290[.0...]	3	M2 for $\frac{6999.31}{\left(1 + \frac{2.16}{100}\right)^5}$ or M1 for $[A]\left(1 + \frac{2.16}{100}\right)^5$
15	73	3	B1 for angle $PBC = 52$ B1 for APO or $BPC = 55$ or APC or $OPB = 125$
16	tangent ruled at $x = 2$	B1	
	-0.7 to -0.3	B2	dep on B1 or a close attempt at tangent at $x = 2$ or M1 for rise/run for their tangent at $x = 2$ must see correct or implied calculation from a drawn tangent
17(a)	-3	1	
17(b)	$\frac{m}{4}$ or $0.25m$ final answer	2	B1 for $\frac{1}{4}$ or 0.25 or 4^{-1} or m correct in final answer
18	917 or 918 or 917.4 to 917.6	3	M2 for $\pi \times 2.6^2 \times 12 \times 60 \times 60 \div 1000$ or M1 for $\pi \times 2.6^2$ isw or $12 \times 60 \times 60 \div 1000$ isw If 0 scored SC1 for figs 917 to 918
19	$\frac{b}{a+b}$ final answer	3	B1 for $b(a-b)$ B1 for $(a+b)(a-b)$
20(a)	$\begin{pmatrix} 7 & 8 \\ -11 & 36 \end{pmatrix}$	2	B1 for 2 correct elements
20(b)	4	2	M1 for $3x - (-1) \times (-7) = 5$ or better

Question	Answer	Marks	Partial Marks
21	$\frac{25}{8}$	B1	or $\frac{75}{24}$
	<i>their</i> $\frac{25}{8} \times \frac{12}{5}$ or <i>their</i> $\frac{75}{24} \div \frac{10}{24}$ oe	M1	$\frac{75}{24} \times \frac{24}{10}$
	<i>their</i> $\frac{300}{40}$ oe	M1	oe e.g. $\frac{1800}{240}$, $\frac{75}{10}$, $\frac{60}{8}$, $\frac{30}{4}$, $\frac{15}{2}$
	$7\frac{1}{2}$ cao	A1	
22(a)	$1\frac{2}{3}$ or 1.67 or 1.666 to 1.667	1	
22(b)	1062.5	3	M2 for $\frac{25}{2}(50+35)$ oe or M1 for one area
23(a)	(4.5, – 1)	2	B1 for each
23(b)	$[y=]\frac{5}{8}x + \frac{7}{4}$	4	M1 for $\frac{-5-3}{7-2}$ oe M1 for –1/ <i>their</i> $-\frac{8}{5}$ M1 for $3 = 2 \times \text{their gradient} + c$ oe
24(a)	5.95 or 5.954...	3	M2 for $\frac{7.4}{\sin 97} \times \sin 53$ or M1 for $\frac{\sin 97}{7.4} = \frac{\sin 53}{SR}$ oe
24(b)	3.73 or 3.733 to 3.734	4	M2 for $8.5^2 + 7.4^2 - 2 \times 8.5 \times 7.4 \times \cos 26$ or M1 for implicit form A1 for 13.9[4...]